

# Scaling Challenges of Semiconductor Packaging in the Era of Big Data

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## Abstract

In advent of multimedia, social media and Internet of Things, our world is exploding with enormous amount of data, so-called Big Data. The use of Big Data provides us with opportunities to bring solutions and innovations to variety of industries such as healthcare, energy, banking and automotive. On the other hand, the computing requirement to analyze this large volume of data is becoming higher than ever. The exascale computing is required in the Era of Big Data. In order to achieve this demand, further technology innovations for package scaling such as 3D-IC with TSV (through silicon via) are needed. The fine pitch die-to-die interconnection is a key element in increasing the total bandwidth in 3D integration. The important technologies in 3D integration include micro-bumping, thermally enhanced underfill materials and advanced interposers. Material selection for reliable fine-pitch interconnection has become a critical challenge in 3D chip stacking. Underfill material in die-to-die device is also a critical element in reducing total packaging stress and in enhancing vertical thermal conductivity. Low CTE high-density organic substrate is an emerging technology for 2.5D structure.

## Key words

3D-IC, Big Data, micro-bump, Underfill

## I. Introduction

2.5 quintillion bytes of data are created everyday and 90% of the data in the world today has been created in the last two years alone. This data comes from everywhere such as sensor data of Internet of things, posts in social media and digital data in multimedia. To analyze this enormous amount of data – so-called Big Data, a much faster rate of improvement in system performance is required. For example, 4x/2years which is the double of the historic trend (2x/2years) is needed to reach exaflop capabilities (100x-1000x over present systems). In order to deliver this extremely higher system-level performance, new technology innovations are needed. Historically, the improvement of system performance was lead by the device scaling and its related innovations. However, the performance of such extremely scaled devices has hit the wall not only due to the physical limit of scaling, but also due to the limited scaling of packaging technologies including interconnections between chips. 3D integration is one of the most promising technology in multi-core architectures for increasing the

total bandwidth and for reducing overall power consumption by reducing interconnect capacitance, allowing integrated voltage regulations.

## II. Key Technologies for 3D-IC

### A. Controlled IMC micro-joints

C4 (Controlled Collapse Chip Connection) solder joining has been successfully used in die to substrate flip chip interconnection whose pitch ranges between 120 $\mu$ m and 150 $\mu$ m. However, the 3D-IC requires less than 50 $\mu$ m pitch die-to-die interconnection and hence the C4 is not suitable for such system. The low-volume solder joining is an attractive method for 3D-IC, since it offers such benefits as extension to fine-pitch integration, increased vertical heat transfer and hierarchy for repeated thermal processes without re-melting. The low-volume solder mostly forms IMCs (intermetallic compounds) with underlying metals. The IMCs are brittle and could cause high mechanical stress in the joints and the stacked structure. High mechanical

stress in 3D-IC potentially leads to the device failures, since the 3D device consists of fragile components such as thin Si chips and through-silicon-vias, which are under CPI (chip-package interaction).

To investigate the package level stress, two metallurgies with different mechanical properties for interconnections between silicon dies were evaluated [1]. One is the Cu/Sn solder joint in which IMC layers are easily created (Fig.1 (a)). The other is the Cu/Ni/In solder joint in which most of the In solder remains unreacted even after bonding due to the presence of the Ni barrier (Fig.1 (b)). Deep thermal cycle (DTC) tests (-55 to +125 °C) of the chip stacks (Fig.2 (a)) and the chip stacks flip-chip attached onto the organic substrates (Fig.2 (b)) were conducted. A higher stress is imposed onto the solder joints in the configuration in Fig.2 (b) than those in the configuration (a) due to the large CTE mismatch between Si dies and the organic substrate.

For the chip stacks on organic substrates showed fails through DTC stress, and the failure modes were different between the Cu/Sn and Cu/Ni/In joints showed different failure modes after DTC test (Fig.3). The failed samples with Cu/Ni/In joints showed cracks inside the In solder joints. However, the failed samples with Cu/Sn joints showed no cracks inside the solder joint, but large cracks inside the top silicon dies for some parts and interfacial cracks between metal pads and Cu-Sn IMCs for some parts.

The chip stacks showed no electrical open fails for both metallurgies, but the average change in the test chain resistance of the chip stack with Cu/Sn joints slightly increased whereas that of the chip stack with Cu/Ni/In joints remained stable (Fig.4). To analyze the joints in detail before and after DTC stressing, the samples of both metallurgies were grinded and polished followed by ion milling. Fig. 5 and 6 show cross-sectional SEM images of the Cu/Sn and Cu/Ni/In joints respectively. For the Cu/Sn joint before DTC stressing, Cu-Sn IMCs connect Cu and Ni metal pads directly and there are some Sn solder portions surrounded by the Cu-Sn IMCs (Fig.5 (a)). The Sn solders were transformed into IMCs under the thermal cycle environment (Fig.5 (b)). The IMC layer between the Ni pad and  $\text{Cu}_3\text{Sn}$  IMC layer is considered to be  $(\text{Au}, \text{Ni}, \text{Cu})_6\text{Sn}_5$  based on the EDX quantitative analysis (table 1). More Cu was consumed after the DTC 3,000 cycles and some kirkendall voids between Cu and  $\text{Cu}_3\text{Sn}$  layers were seen. However no cracks were found inside the Cu/Sn joints of configuration (a) chip stacks. For the Cu/Ni/In joint, the In solder layer exists before DTC stressing (Fig.6 (a)). However, the In solder layer was transformed into IMCs under the thermal cycle environment (Fig.6 (b)). In the Cu/Ni/In joints after DTC 3,000 cycles, IMCs of Ni-In or Cu-Ni-In (L5, L7, L8) were formed at the Ni pad sides, and the IMC considered to be  $\text{AnIn}_2$  (L6) from the analysis was formed between the Ni-In

(or Cu-Ni-In) IMCs. There was a line observed between L5 and L6 layers which looks like a crack, but a resistance increase was not seen in the sample as shown in Fig.4.

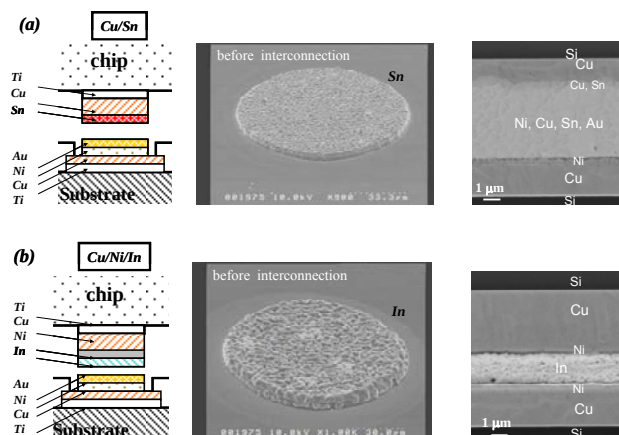


Fig.1 Schematic illustrations of joint with (a) Cu/Sn bump and (b) Cu/Ni/In bump.

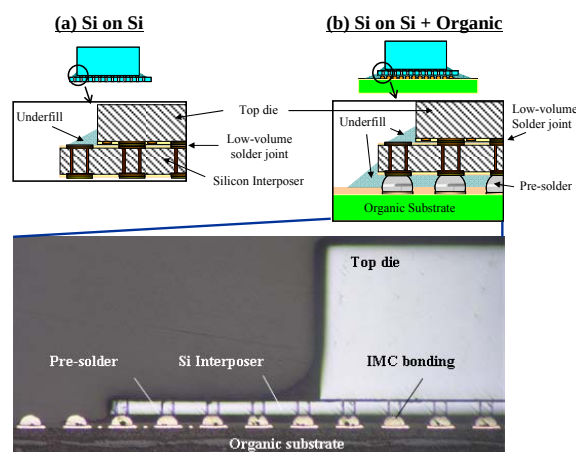


Fig.2 Configurations of test vehicles (a) without an organic substrate and (b) with an organic substrate.

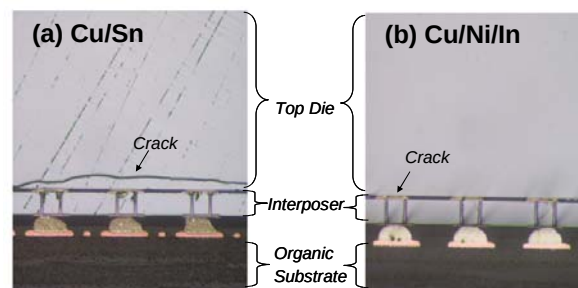


Fig.3 Cross-sectional micrograph of Si/Si on organic substrate samples showing (a) a crack inside the top Si dies in typical Cu/Sn joints and (b) a crack inside joints in typical Cu/Ni/In joints

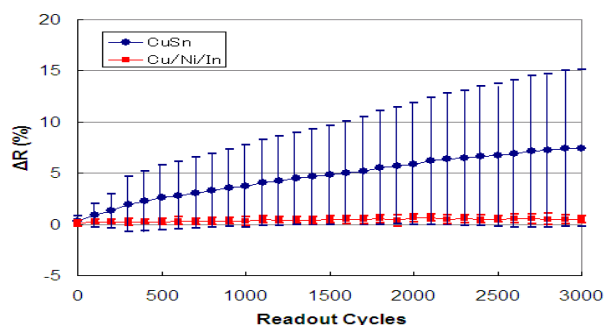


Fig.4 The average values of the change in resistance, from the initial value for the the Si/Si systems with Cu/Sn and Cu/Ni/In joints for DTC.

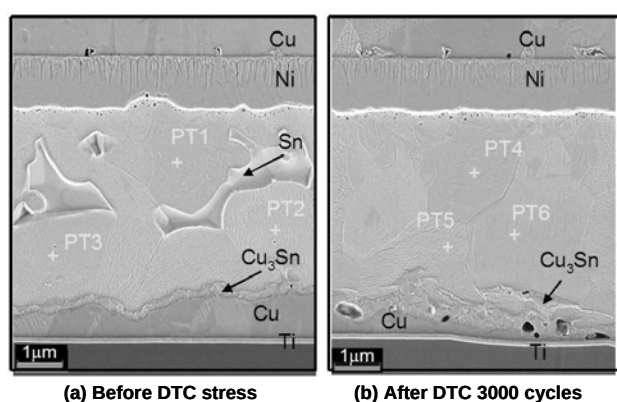


Fig.5 Si on Si with Sn/Cu joint

Table 1 EDX quantitative analysis of IMC joints with Cu/Sn bump before and after DTC stress

| Sample     | Analysis | Normalized at. % |    |    |    |
|------------|----------|------------------|----|----|----|
|            |          | Cu               | Sn | Ni | Au |
| Before DTC | PT 1     | 43               | 46 | 5  | 6  |
|            | PT 2     | 49               | 44 | 2  | 5  |
|            | PT 3     | 44               | 45 | 4  | 7  |
| After DTC  | PT 4     | 46               | 47 | 4  | 3  |
|            | PT 5     | 45               | 48 | 4  | 3  |
|            | PT 6     | 47               | 47 | 3  | 3  |

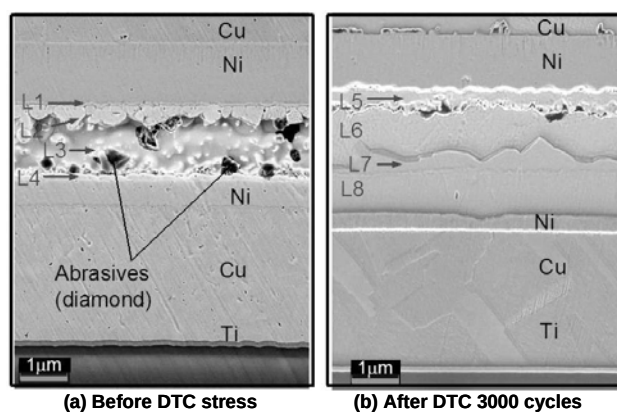


Fig.6 Si on Si with Cu/Ni/In joint

Mechanical properties of the IMCs formed for Cu/Sn and Cu/Ni/In joints are different. The yield stress of the CuSn IMC (1,230 MPa [2]) is greater than that of the AuIn<sub>2</sub> IMC (2-230 MPa estimated from the shear strength and microhardness in [3][4]). The yield stress of CuSn IMC is comparable or higher than that of silicon dies which generally ranges between 300-1,000MPa.

The higher yield stress of CuSn IMC than silicon dies can explain the formation of cracks in the silicon die or at the silicon/underfill interfaces, rather than in the Cu-Sn IMC joint stacks for the die stacks on organic substrates with Cu/Sn joints. The die stacks with Cu/Ni/In joints showed cracks inside the joint likely, because the yield stress of AuIn<sub>2</sub> IMC is lower than that of silicon dies. The observed increase in daisy chain resistance during the DTC test for the die stacks with Cu/Sn joints may also be associated with the crack formation at the sites other than inside the Cu-Sn IMC joints. Even though, in the die stacks without organic substrates, the joint stress is much smaller than that for the die stacks on organic substrates.

One of the possibilities for the resistance increase is a resistivity change due to the IMC formation. Calculated resistance change by assuming the transformation from the relatively lower resistivity Cu and Sn joint to the higher resistivity Cu<sub>3</sub>Sn and Cu<sub>6</sub>Sn<sub>5</sub> joint is less than 0.1%. Therefore it is unlikely that the IMC formation and Cu consumption in the joint are leading to a resistance increase of the joints.

The finite element analyses (FEA) were performed on the die stack on organic substrate in order to understand the failures observed in experiments. The analyses were conducted using the sub-modeling technique. The 1/4 model was used for the analysis considering the symmetry of the structure (Fig.7). The 1/4 macro model consists of a 725μm thick top silicon die, silicon interposer with W-TSVs, a C4 bump layer and an organic substrate. Composite material properties were used for the C4 layer, Si interposer with W-TSVs and the micro-bumps with underfill layer. The composite material properties were obtained using a commercial software, ANSYS MultiScale Sim. For the Cu/Sn bump and the Cu/Ni/In bump, the material properties of CuSn IMC and AuIn IMC were assumed. The material properties used in the analyses are listed in Table 2. The bump pitch is 200μm, the bump diameter is 100μm and the diameter of TSVs is 60μm. The micro-model consists of metal pads with IMC joints as shown in Fig. 7. The analyses were conducted from 125°C to -55°C and assumed zero stress at 125°C. A cross-section of the stress contour and the contour of the plastic strain is shown in Fig. 8. The maximum value of the equivalent plastic strain for AuIn<sub>2</sub> IMC was several order of magnitude higher (0.018) than that for CuSn IMC (0.9e-4). It is consistent with the crack in the

joint observed for the die stacks on organic substrate with Cu/Ni/In joints. For the die stacks with Cu/Sn joints, the maximum value of the first principal stress on top silicon die was 300MPa, which is comparable to the yield stress of the silicon. The analysis is consistent with the formation of cracks inside the silicon dies for the die stacks on organic substrate with Cu/Sn joints. The stress due to the CPI on the silicon die may have exceeded the yield stress of the silicon die.

In addition to the CPI challenge, the electromigration has emerged as a critical reliability concern because the joint size in 3D-ICs is significantly smaller than that for the C4 solder joints. There is a great demand in increasing the current density. Since the solder volume is substantially reduced due to the fine-pitch requirements, the limited supply of Sn induces more extensive IMC formation and can convert the entire bump into IMC upon thermal annealing or under current stressing. However, we showed that this IMC formation plays an important role in improving the electromigration reliability of micro-bumps (Fig.9) [6].

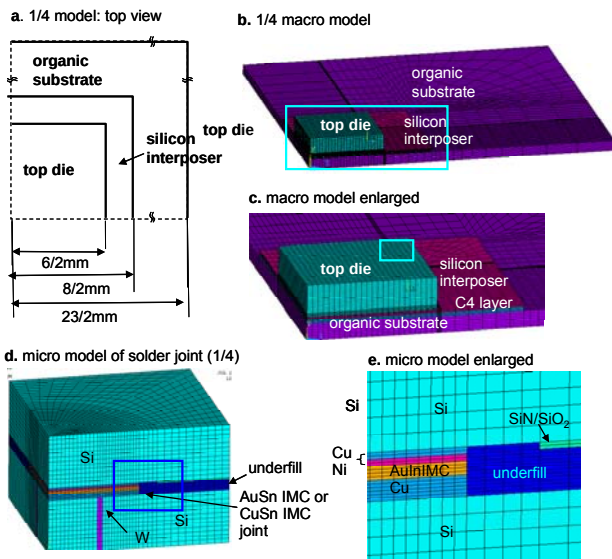


Fig.7 FEA Model for die stack on an organic substrate

Table 2 Material's properties used in FEM (Finite Element Modeling)

| Material          | Modulus (GPa) | CTE (ppm/K) | Poisson's Ratio | Yield stress (MPa) | Ref.   |
|-------------------|---------------|-------------|-----------------|--------------------|--------|
| Silicon           | 160           | 2.56        | 0.23            | -                  |        |
| Copper            | 117           | 16.5        | 0.3             | 69                 |        |
| AuIn <sub>2</sub> | 21            | 24          | 0.4             | 15                 | [3][4] |
| CuSnIMC           | 110           | 23          | 0.3             | 1230               | [2][5] |
| Tungsten          | 360           | 4.5         | 0.3             | -                  |        |

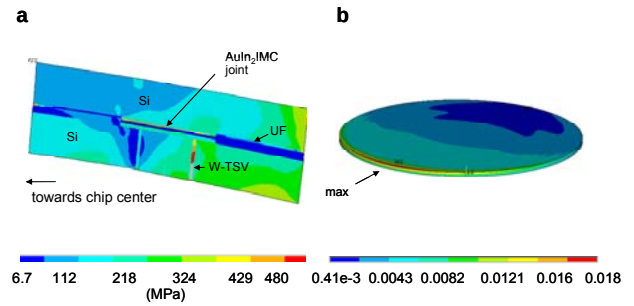


Fig.8 FEA result of die stack on organic substrate with Cu/Ni/In joints (a) von Mises stress contour of the Cu/Ni/In joint (b) equivalent strain of the AuIn<sub>2</sub> IMC

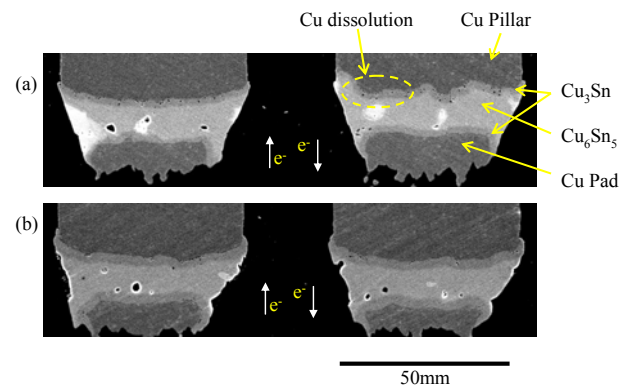


Fig.9 Cross-sectional SEM images of solder capped Cu pillar joint after the electromigration test at 125°C 10kA/cm<sup>2</sup> 1,000h (a) Without aging (Cu<sub>3</sub>Sn t0.5µm) (b) Aging at 200°C 10h (Cu<sub>3</sub>Sn t1.5µm).

## B. Thermally Enhanced Underfill Material

Thermal modeling is used to understand the heat flows and the effectiveness of various thermal underfill materials for 3D stacks. Fig.10 shows a joint layer consisting of solder bumps and a standard underfill material (0.4W/mK). The joint layer is a bottleneck of the thermal resistance in this packaging structure [7]. The simulation result shows that if a high thermal conducting underfill material (2.0W/mK) is used, the thermal resistance is dramatically reduced. A common method to enhance the thermal conductivity of underfill materials is to increase its filler contents. In 3D fine-pitch systems, the high filler loading materials can not be applied using the standard capillary underfill technology due to narrow gaps between Si chips. The pre-applied underfill technology is developed to overcome this issue. The OBAR (Over-Bump Applied Resin) is one of the pre-applied processes [8], in which an epoxy resin is applied over the bumps of a wafer and b-staged. The b-staged wafer is diced and OBAR coated die is aligned to a



substrate and joined by melting the solder bumps by flip chip bonder. Pre-applied underfill on substrate has been used for consumer products, but one of the challenges is to avoid inclusions of filler particles in the solder joint, since they can affect the reliability of these connections.

3D chip stacking process was performed by using pre-applied thermal underfill [9]. The test vehicle consists of stacked dies (10mm×10mm×0.1mm), a silicon interposer (20mm×20mm×0.2mm), and organic substrate (40mm×40mm×0.76mm). All silicon dies have 150μm pitch area array Cu-pillar bumps (30μm high), no TSV, and no structures on backside of the dies. As shown by a red line in Fig.11, all 3,844 bumps between the bottommost die and the silicon substrate can be connected electrically by daisy chains.

In the assembly process, the thermally enhanced underfill was dispensed on each 10mm dies by a dispenser at first. The dispensed volume was precisely controlled to make appropriate fillet shape and volume. Then, first three dies were stacked sequentially on the silicon substrate at 100°C by a flip-chip bonder. At this moment, lead-free solder capped on the Cu-pillar has not been melted yet. When the top die was mounted on the stacked dies, high temperature (250°C) was applied to make the solder joints. This process flow was developed by IBM to eliminate excess thermal history for 3D chip stack modules. It could enable pressure cure process for pre-applied underfill on multi-die-stack 3D modules with low cost. [10] Subsequently, the stacked silicon chip was bonded on the organic substrate with an adhesive containing 50μm spacers to hold an appropriate gap between the silicon substrate and the organic substrate. Finally, a capillary underfill was filled the bottom gap, and cured at 150°C, 4 hours. We successfully confirmed electrical connections of all the bumps on the daisy chains. (Fig.12)

To achieve 2.0W/mK thermal conductivity of underfill materials, standard silica fillers should be replaced by thermally high conductive fillers, such as boron nitride fillers. Since the properties of thermally high conductive fillers are different from the standard fillers, the formulation of thermal underfill materials must be carefully optimized by considering all of the material properties of the compound. In addition, the pre-applied underfill material must act as a soldering flux to create good solder joints. In 3D structures, the silicon dies with fine-pitch TSVs are much thinner than commonly-used dies. Therefore, the underfilling process must have strong gap filling ability and ability to control the volume of materials precisely in order to create appropriate fillet shape with thin dies without contaminating the backside of the die. Thermally enhanced underfill materials with high filler-loading generally show higher viscosities than typical capillary underfill materials, which makes the

material formulation a key challenge.

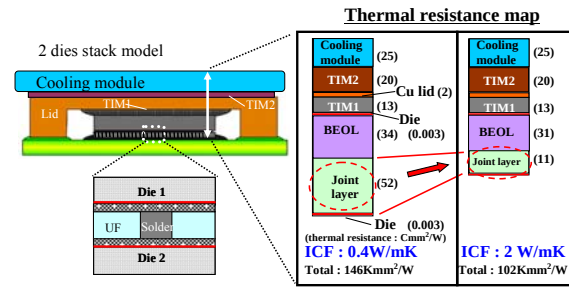


Fig.10 Effectiveness of high thermal conductive underfill.

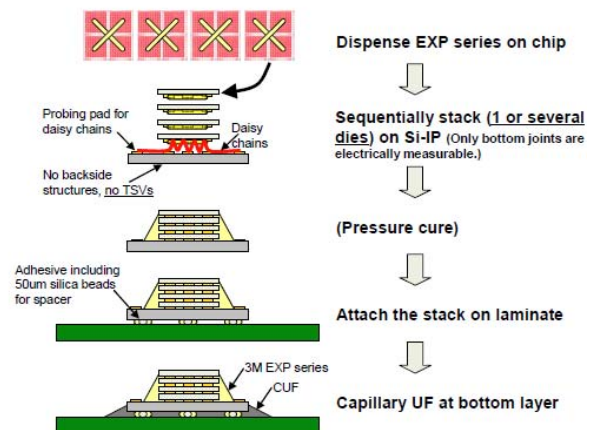


Fig.11 3D Chip Stack Process Flow

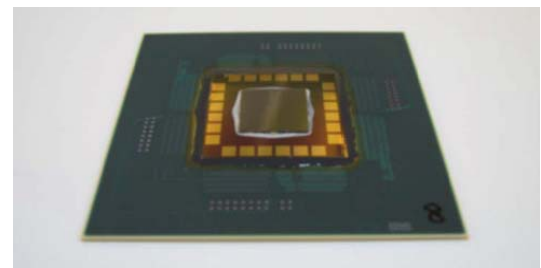


Fig.12 Five dies stack on the organic substrate by Stack Joining process (four dies: 10mm×10mm×0.1mm, one Si-substrate: 20mm×20mm×0.2mm, one substrate: 40mm×40mm×0.76mm) (All 3,844 bumps between bottom die and Si-substrate are electrically connected.)

### C. Advanced Interposer

There is a large gap in design rule between silicon devices and packaging substrates. A typical design rule of BEOL is less than 1μm/1μm, while that of organic substrate is about 15μm/15μm. Therefore, the bump pitch of the silicon device has been kept as wide as 120μm-150μm in order to match the rough design rule of organic substrate.

There are two options to connect Si dies using 2D technology. One is that one die is put on one substrate as

shown in Fig.13 (a).

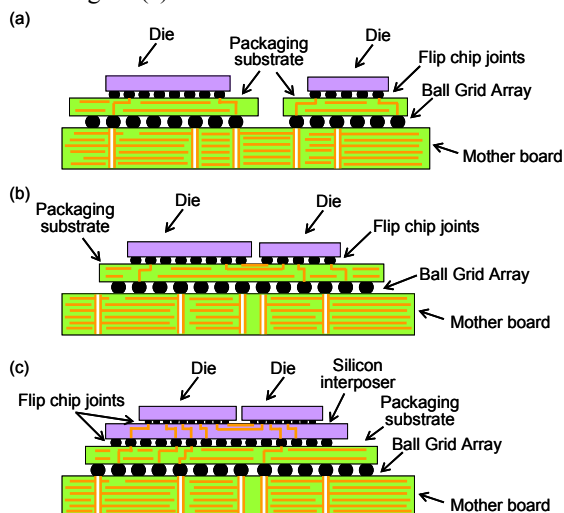


Fig.13 Packaging structures (a) conventional 2D package (b) MCM 2D package (c) 2.5D package.

In this case, the interconnection goes through the BGA whose pitch is about 1mm. The other is that all the dies are put on one substrate to form a MCM (Multi Chip Module) as shown in Fig.13 (b). In this case, the interconnection goes through the flip chip joints whose pitch is about 120 $\mu$ m-150 $\mu$ m. In both cases, the interconnection goes through a long distance which would limit the electrical performance of the system.

To solve the problem, a 2.5D package is proposed as shown in Fig.13 (c). A silicon interposer is used to fill the gap in design rule between the silicon device and the organic substrate in the 2.5D package. A silicon device whose bump pitch is less than 50 $\mu$ m can be assembled on the silicon interposer when the silicon interposer is made using BEOL process. However, the size of the silicon interposer would be large because multi-dies are put on the silicon interposer, which would cause a large stress due to the CTE mismatch between the silicon interposer and the organic substrate.

To support the large size of silicon interposer, an extremely low CTE organic substrate is expected for 2.5D package. By using the substrate made by extremely low CTE material, the stress in solder joints caused by the large CTE mismatch between Si dies and the substrate can be alleviated.

Furthermore, if the design rules of organic substrates become as fine as those for silicon interposers, they can replace the silicon interposers, which could lead to a large cost reduction.

### III. Conclusion

Key technologies in 3D integration, namely

micro-bumping, thermally enhanced underfill, advanced interposer and interchip underfill materials are discussed. The optimization of the metallurgies and the structures of the micro-joints are the keys to reduce thermo-mechanical stress caused by CPI between Si dies and organic substrates. The IMC formation in fine-pitch interconnections has shown to be effective in improving the electromigration reliability. Thermal simulation showed the importance of the role of underfill layers as thermal conducting agents in 3D die stacks. Thermally enhanced underfill materials has been developed using high-conductive fillers such as boron nitride fillers. Optimization of underfilling process as well as material optimization were performed to overcome the challenges in filling the narrow gaps between thin Si dies in 3D structures. The development of extremely low-CTE organic substrate can lead to a reduction of stress in solder joints caused by the large CTE mismatch between Si dies and the substrate, which is a critical concern for high-end server applications in which large Si dies are being used. There are tremendous potentials in the use of Big Data in creating innovations and in solving various problems in our world. In order to expand the Big Data technologies further, the advancement of the computing technologies is vital, such as bringing exascale computer into realization. 3D integration is one of the most promising technology for the purpose and closing the gap between the Si device scaling and packaging scaling is a key for the future advancement of the computing technology.

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